MILITARY SPECIFICATION

ALARM DETECTOR ID-1974( )/PPS-5

This specification is approved for use by the U.S. Army Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for the construction and testing of an automatic alarm device for use with Radar Set AN/PPS-5( ).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

NN-P-71	Pallets, Material Handling, Wood Stringer Construction, 2-Way and 4-way (Partial)
QQ-S-781	Strapping, Steel, and Seals
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Locked Corner
PPP-B-636	Boxes, Shipping, Fiberboard

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PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-S-760	Strapping, Nonmetallic (and Connectors)
PPP-T-97	Tape, Packaging/Industrial, Filament Reinforced

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-P-11268	Parts, Materials and Processes Used In Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded Bead)
MIL-P-26514	Polyurethane Foam, Rigid or Flexible, for Packaging
MIL-C-55442	Cable Assemblies and Cord Assemblies, Packaging of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads for 40" x 48" Pallets
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects

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MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics Measurements of
MIL-STD-781	Reliability Test Exponential Distribution
MIL-STD-1188	Commercial Packaging of Supplies and Equipment
MIL-STD-1562	Lists of Standard Microcircuits

DRAWINGS

Electronics Command

DL-SM-B-704800	Detector Set Assembly
DL-SM-B-704804	Cable Assembly, Special Purpose Electrical
DL-SM-B-704806	Alarm Detector ID-1974( )/PPS-5
SM-A-704811	Test Procedure, (System Performance Test) Alarm Detector
SM-A-704847	Test Procedure, Environmental Alarm Detector ID-1974( )/PPS-5
SM-A-704849	Test Procedure, Performance Alarm Detector ID-1974( )/PPS-5
SM-A-704853	Test Procedure, Alignment Alarm Detector ID-1974( )/PPS-5

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

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3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification and Drawing DL-SM-B-704800 (including the requirements for parts, materials, and processes thereon).

3.2.1 Major components list. Alarm Detector ID-1974( )/PPS-5 shall consist of the major components listed in Table I.

Table I - Major Components

Component	Quantity	Drawing
Alarm Detector ID-1975( )/PPS-5	1 ea	DL-SM-B-704806
Cable Assembly, Special Purpose, Electrical	1 ea	DL-SM-B-704804

3.3 Parts, materials and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply. Microcircuits shall be selected in accordance with MIL-STD-1562.

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3.4 Performance.

3.4.1 Alarm detector performance.

3.4.1.1 Power requirement. The alarm detector shall require no more than 0.250 watts input power. (See 4.10.1).

3.4.1.2 Input signal level. The alarm detector shall require no more than 10 millivolts of signal. (See 4.10.1).

3.4.1.3 Adaptive highpass filter slope. The alarm detector shall exhibit a frequency response characteristic of at least 24 dB/octave. (See 4.10.1).

3.4.1.4 Tracking range of adaptive cutoff. The tracking range of the adaptive filter shall be from 35 Hz to 350 Hz as measured at the 3 dB power level. (See 4.10.1).

3.4.1.5 Low pass filter slopes (two channels). The alarm detector shall have two velocity channels with the following filter characteristics: (See 4.10.1).

a. Low Velocity Channel

6 dB/octave slope (minimum)
Cut-off at 250 Hz (3 dB power level)

b. High Velocity Channel

18 dB/octave slope (minimum)
Cut-off at 1680 Hz (3 dB power level)

3.4.1.6 Automatic gain control range. The alarm detector shall have an AGC capability covering at least 30 dB. (See 4.10.1).

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3.4.2 System performance. When used with a modified Radar Set AN/PPS-5( ), the alarm detector shall maintain a low constant false alarm rate not exceeding one (1) false alarm in a 1-hour period when the control indicator unit is utilized. This performance shall be exhibited, with a 90% probability of detection, in detecting moving targets under the following conditions: (See 4.10.2).

Condition A

Target:	Walking man
Target velocity:	1 to 3 miles/hr (radial)
Target range:	50 meters to 5000 meters
Wind velocity:	5 to 20 miles/hr
Wind direction:	Radial ($\pm 15^{o}$)
Clutter:	Approximately 50% of resolution cell filled with wind blown foliage.

Condition B

Target:	Moving vehicle
Target velocity:	35 miles/hr (radial) maximum
Target range:	To 10,000 meters
Wind velocity:	10 to 35 miles/hr
Wind direction:	Same as Condition A
Clutter:	Same as Condition A

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3.5 Service conditions. The equipment shall meet the following service conditions. (See 4.8).

3.5.1 Temperature.

3.5.1.1 Low. The equipment shall be capable of operating at -50°F. (A warm up period of 2 minutes maximum is acceptable). The equipment shall be capable of being stored at -70°F without damage or degradation.

3.5.1.2 High. The equipment shall be capable of operating at +160°F.

3.5.2 Thermal shock. The equipment shall be capable of operating after being subjected to the thermal shock test.

3.5.3 Rain. The equipment shall be capable of operating during and after exposure to rain as determined by the rain test.

3.5.4 Altitude. The equipment shall be capable of operating at 10,000 feet altitude and being exposed to 50,000 feet without degradation.

3.5.5 Leakage. The equipment shall be submersion proof to a covering depth of three (3) feet of water for not less than two (2) hours, with no evidence of leakage.

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3.5.6 Dust. The equipment shall be capable of operation after being subjected to the dust test.

3.5.7 Humidity. The equipment shall exhibit no physical damage, such as corrosion, rust, blistering, swelling or deterioration of parts and materials after being subjected to the humidity test.

3.5.8 Fungus. The equipment shall withstand in the operating mode, but nonoperating condition, exposure to fungus growth as encountered in tropical climates and shall exhibit no visible evidence of fungus.

3.5.9 Salt fog. The equipment shall withstand in an operating mode, but nonoperating condition, exposure to a salt fog atmosphere. There shall be no evidence of degradation, such as flaking, pitting, blistering or loosening of finish or metal surface. Electrical operation is not required.

3.5.10 Shock, bench handling. The equipment shall withstand shocks which may be encountered during bench servicing.

3.5.11 Vibration. The equipment shall meet specified performance requirements after being subjected to the loose cargo bounce test.

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3.5.12 Shock, drop. The alarm detector shall meet specified performance requirements after being subjected to the drop test.

3.6 Interchangeability. Like units, assemblies, sub-assemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454. (See 4.7).

3.7 Electromagnetic compatibility. The generation of electromagnetic interference (EMI) by the alarm device and its susceptibility thereto shall be in compliance with the requirements as specified in MIL-STD-461, Notice 4. (See 4.9).

3.7.1 Electromagnetic compatibility requirement. The alarm detector shall meet the emission and susceptibility requirements of Table VI, Ground CE Equipment and Component Items of Equipment, of MIL-STD-461, Notice 4 as defined herein for nonsheltered equipments.

(1)(2) CE01	(1)(3) CS01	(1)(3) RS03
(2)(2) CE03	(1)(3) CS02	(1)(3) RS03.1
(3)(2) CE04	(1)(3) CS06	

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(1) Tests shall be performed using DC external power as specified in 3.4.1.1 and Figure 1.

(2) Conducted and radiated switching transients shall comply with the limit of paragraphs 6.2.1 and 6.13.4 of MIL-STD-461, Notice 4.

(3) During susceptibility testing, degradation of performance shall be defined as following:

a. The input signal level shall be adjusted to 3 dB below the sensitivity threshold. The alarm detector is susceptible if the alarm comes on.

b. The input signal level shall be adjusted to 3 dB above the sensitivity threshold. The alarm is susceptible if the alarm ceases.

c. If the signal appearing at the headset terminals varies more than ±3 dB when the susceptibility signal is applied, the alarm detector is susceptible.

3.8 Safety engineering.

3.8.1 Personnel hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly processes, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual examination and inspection. (See 4.13).

3.8.2 Edge and corner rounding. All exposed edges and corners shall be rounded to eliminate personnel hazards.

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3.8.3 Radioactive materials. Radioactive materials shall not be used.

3.9 Finish. The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings. (See 4.4).

3.10 Marking. (See 4.4).

3.10.1 General. Marking shall conform to MIL-M-13231. Exterior marking shall be group I as described in that specification.

3.10.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.10.3 Serial numbers. Each Alarm Detector ID-1974( )/PPS-5 shall be serial numbered.

3.11 Preconditioning. Each equipment shall meet the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.

3.12 Burn-in. Each equipment shall meet the requirements herein, without subsequent processing, after subjection to the burn-in of 4.11.

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3.13 Workmanship. The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs of MIL-P-11268: (See 4.13).

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
  Slack.
  Protection.
  Clearance.
  Splicing.
  Connections, general.
  Grounding, general.
  Shielding on wire and cable.
Soldering and brazing.
Cleaning.
General requirements for securing of parts.
Cleaning prior to welding.
Process for welding.
Cleaning after welding.
Controls.
Self-tapping screws.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Test equipment and facilities for Government Inspection shall be made available by the contractor.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.3) - does not include packaging.

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before packaging. (See 4.5).

(2) Quality conformance inspection of packaging. (See 4.12).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

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4.3.1 First article units. The contractor shall furnish five (5) first article units of the complete Alarm Indicator ID-1974( )/PPS-5.

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in Table II and shall be performed in the order specified therein.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

Table II - First Article Inspection

Inspection	Reqt. Para.	Test Para.	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5
1. Inspections covered by subsidiary documents		4.4	Inspections to be performed on all units				
2. Bounce preconditioning	3.11	4.6	Inspections to be performed on all units				
3. Burn-in	3.12	4.11	Inspections to be performed on all units				
4. Group A inspection	(See Table III)		Inspections to be performed on all units				
5. Group B inspection	(See Table IV)		Inspections to be performed on all units				

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Table II - First Article Inspection (Cont'd)

Inspection	Reqt. Para.	Test Para.	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5
6. System performance	3.4.2	4.10.2	1	7			
7. EMI	3.7	4.9			1		
8. Environmental	3.5	4.8					
Low Temperature	3.5.1.1		2				
High Temperature	3.5.1.2		3				
Thermal Shock	3.5.2			1			
Rain	3.5.3			2			
Altitude	3.5.4			3			
Leakage	3.5.5		5,7				
Dust	3.5.6			4			
Humidity	3.5.7						1
Fungus	3.5.8					1	
Salt Fog	3.5.9		4				
Shock, bench handling	3.5.10			5			
Vibration	3.5.11			6			
Shock, drop	3.5.12		6				

NOTES:

a. The inspections 1 through 5, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

b. The numbers in the unit columns in the table covered by inspections 6 through 8 specify the order of inspections for the indicated unit.

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c. The equipment shall be thoroughly washed, cleaned, dried and refurbished as necessary after the fungus and salt fog tests before proceeding with subsequent inspection.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection requested by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Requirement
Parts, materials and processes	3.3
Finish	3.9
Marking	3.10

4.5 Quality conformance inspection of equipment before packaging. Every equipment shall be subjected to the preconditioning burn-in procedure of 4.11 prior to conformance inspection. The contractor shall perform the inspection specified in 4.4, 4.6, and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirements as a result of this preconditioning shall be classified as a defect and recorded as such.

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4.5.1 Group A inspection. Each unit on contract or order shall be inspected for conformance to the inspections specified in Table III. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and the AQLs indicated in Table III.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

Table III - Group A Inspection

Inspection or Test	Reqt. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and Mechanical				
Alarm Detector ID-1974( )/PPS-5	3.13	4.13	1.0%	4.0%
Cable Assembly, Special Purpose, Electrical	3.13	4.13	1.0%	4.0%
Operational and Electrical Tests				
Alarm Detector Performance	3.4.1	4.10.1	1.0%	*

* All defects shall be considered major.

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4.5.2 Group B inspection. This inspection shall conform to Table IV and the special inspection levels of MIL-STD-105. Group B inspection shall normally be performed on inspection lots that have been subjected to and met Group A inspection.

4.5.2.1 Group B sampling plans. The Group B sampling plans, for the AQLs listed in Table IV shall be inspection level S-4.

Table IV - Group B Inspection

Inspection Or Test	Reqt. Para.	Insp. Para.	AQL
Interchangeability	3.6	4.7	6.5%

4.5.3 Group C inspection. This inspection shall consist of the tests specified in Table V and shall be performed on sample units that have been subjected to and met Group A and Group B inspection. Sample units shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for inspection. Four (4) sample units; one for each inspection subgroup cited in Table V, shall be selected at random from every 250 units or fraction thereof produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot. The salt fog and fungus inspections shall be performed only on the first sample units selected for the first Group C inspection.

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4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in the order shown in each subgroup in Table V.

4.5.3.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract. (See 6.3).

Table V - Group C Inspection

Inspection	Reqt. Para.	Insp. Para.
Subgroup 1		
1. System performance	3.4.2	4.10.2
2. Leakage	3.5.5	4.8
3. Shock, drop	3.5.12	4.8
4. Leakage	3.5.5	4.8
Subgroup 2		
1. EMI	3.7	4.9
2. Altitude	3.5.4	4.8
3. Salt fog*	3.5.9	4.8
Subgroup 3		
1. Low temperature	3.5.1.1	4.8
2. High temperature	3.5.1.2	4.8
3. Thermal shock	3.5.2	4.8
4. Rain	3.5.3	4.8
5. Humidity	3.5.7	4.8

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Table V - Group C Inspection (cont'd.)

Inspection	Reqt. Para.	Insp. Para.
Subgroup 4		
1. Dust	3.5.6	4.8
2. Vibration	3.5.11	4.8
3. Shock, bench handling	3.5.10	4.8
4. Fungus*	3.5.8	4.8

*The equipment shall be thoroughly washed, cleaned, and dried after this inspection before proceeding with subsequent inspection.

4.5.4 Reinspection of conforming Group B and Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Bounce preconditioning. The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to inspection.

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4.7 Inspection for dimensional interchangeability. In addition to Requirement 7 of MIL-STD-454, the dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.6. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers and insertable assemblies, when such dimensions affect mating or parts.

b. Dimensions of cavities, when such dimensions effect insertion of items.

c. Location of hinges and fasteners on separable parts of assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major units.

e. Size and form of special threads.

f. Dimensions of major components.

4.8 Service condition tests. Service condition tests shall be performed as detailed in SM-A-704847. Each unit subjected to the service condition tests shall have passed Group A tests and shall be resubmitted to and passed Group A tests after completion of all service condition tests. Where operational tests are specified during service condition cycling, the conditions specified in SM-A-704847 shall be met. These same conditions shall be met at the conclusion of each separate service condition test. (See 3.5).

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4.9 Electromagnetic interference tests. The alarm detector shall be tested to determine compliance with the EMI requirements specified in 3.7.1. Operational mode shall be as indicated in Figure 1. Test methods and procedures shall be in accordance with MIL-STD-462, Notice 3 as implemented by a contractor prepared, Government approved EMI test plan. Testing shall be performed using interconnecting cables of the same type and lengths as will be used in the actual installation.

4.10 Performance tests.

4.10.1 Alarm detector performance tests. Tests for each alarm detector shall be conducted in the laboratory with laboratory instruments and personnel. Tests shall be performed in accordance with the procedure of SM-A-704849 to determine compliance with the following performance requirements. The alignment procedure of SM-A-704853 shall apply.

Performance requirement	Ref Para.
Power requirement	3.4.1.1
Input signal level	3.4.1.2
Adaptive high pass filter slopes	3.4.1.3
Tracking range of adaptive cut-off	3.4.1.4
Low pass filter slopes	3.4.1.5
AGC range	3.4.1.6

4.10.2 System performance tests. Field tests shall be conducted at test site(s) after installation of the alarm detector in the modified AN/PPS-5 radar set. System performance tests shall be performed in accordance with the procedure of SM-A-704811 to determine compliance with the system performance requirements of 3.4.2.

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4.10.2.1 Field test monitoring. An umpire shall be used for all field tests. He shall verify the azimuth and range of the radar when the test item alarms on a moving target.

4.10.2.2 Definition of target terms. Target terms shall be construed to have the following meanings:

a. Actual targets. Targets whose physical presence can be confirmed by the umpire by either visual sighting or radar aural indication, or both.

b. Declared targets. Targets detected by the test item and confirmed by the umpire.

c. Missed targets. Targets confirmed by the umpire but are not detected by the test item.

d. False alarms. Targets detected by the test item but not confirmed by the umpire.

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4.11 Burn-in. Prior to submission to the specified quality conformance inspections, each unit shall be operated for a minimum of 48 hours under the conditions specified in Test Level C, including voltage cycling, of MIL-STD-781. The burn-in shall be concluded by a failure free period, the duration of which shall be 24 hours. Paragraphs 5.1.5 and 5.10 inclusive of MIL-STD-781 shall apply. The types of failures and required actions shall be in accordance with paragraph 5.5.1 of MIL-STD-781. Determination of compliance shall be in accordance with paragraph 5.4.8 of MIL-STD-781.

4.11.1 Monitoring of burn-in. Equipment performance cited in paragraph 4.2 of SM-A-704847 shall be monitored at least once during each equipment "on" cycle. The equipment "off" cycle shall not exceed one (1) minute.

4.11.2 Failure definition. A failure is defined as the inability of the equipment to perform its required function. This includes catastrophic failures as well as any condition which causes departure from performance as required herein.

4.12 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine compliance with the requirements of Section 5 herein:

4.13 Visual and mechanical inspection. The equipment shall be examined for the defects listed in MIL-STD-252. In addition, the equipment shall be visually inspected to determine that the requirements of 3.8 (safety engineering) are met.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or commercial, as specified in the contract or order. (See 6.2b).

5.1.1 Level A.

5.1.1.1 Cleaning. Each Alarm Detector ID-1974( )/PPS-5 shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each Alarm Detector ID-1974( )/PPS-5 shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. Not required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be individually unit packed method IC-1.

5.1.1.4.2 Cable assembly. Each cable assembly shall be individually unit packed in accordance with MIL-C-55442.

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5.1.1.4.3 Alarm Detector ID-1974( )/PPS-5. Each detector shall be individually unit packed method III as follows: Position the captive protector caps over all connectors of the device. Cushion the device together with the cable assembly unit packed as specified in 5.1.1.4.2 on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed to protect the unit from shock and vibration hazards during handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. The cushioned item shall be inclosed within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B or commercial, as specified in the contract or order. (See 6.2b). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

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5.2.1 Level A. A quantity of Alarm Detector ID-1974( )/PPS-5, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, finish A.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.2 Level B. A quantity of Alarm Detector ID-1974( )/PPS-5, unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds,

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containers shall be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A quantity of Alarm Detector ID-1974( )/PPS-5, unit packed as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-1188.

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6. NOTES

6.1 Intended use. The Alarm Detector provides reliable automatic target detection performance by Radar Set AN/PPS-5( ) when the radar is being utilized in a high clutter environment.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A, level B or Commercial preservation and packing. (See section 5).

c. When rough handling test is required.

d. First article test plan and test report(s). (See 4.3.3).

e. Place of final inspection.

f. Technical literature required.

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6.3 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. For details, see the applicable clause of the contract. (See 4.5.3.3).

6.4 Nomenclature. The parentheses in the nomeclature will be deleted or replaced by a letter identifying the particular design; for example: Alarm Detector ID-1974W/PPS-5. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

6.5 Environmental pollution prevention measures for packaging material. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

CUSTODIAN:
Army EL

Preparing Activity:
Army EL
Project No. 5840-A-186

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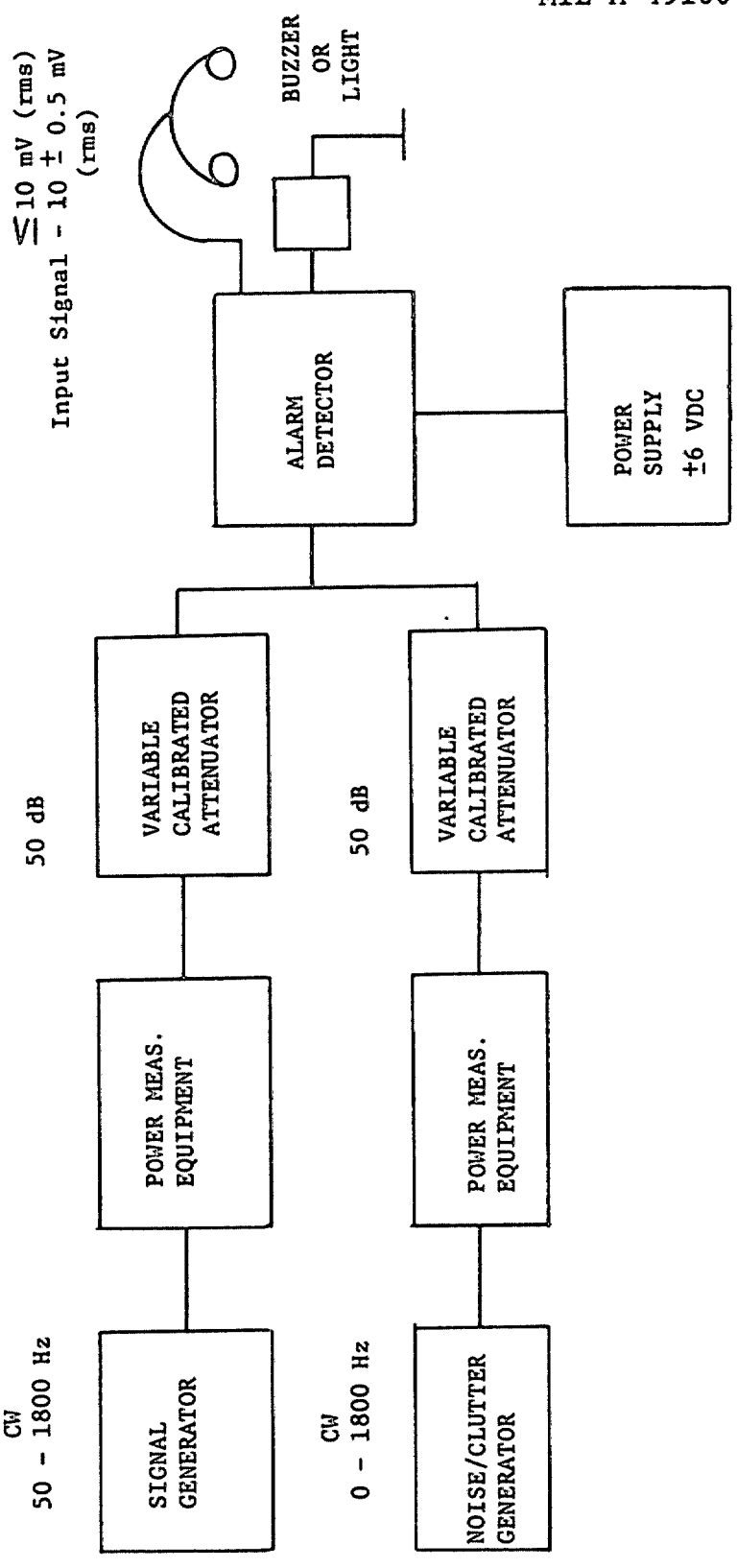


FIGURE 1. BLOCK DIAGRAM EMI TEST SET UP

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22 R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-A-49180(EL) ALARM DETECTOR ID-1974( )/PPS-5

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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